SIMATS ENGINEERING
ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME: COMPUTER VISION **COURSE CODE:ITA0518**

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics.* (BTL 5)

Assessment Tool Weightage:20%

Analyze the given problem by comparing available techniques against the specified constraints and requirements. Evaluate and justify your decision using logical reasoning, clearly indicating how the chosen approach satisfies the conditions.

2. Autonomous Vehicle Detection Logic
A vehicle detection system must:

- Prioritize safety (accuracy > 90%)
- Maintain response time < 50 ms

Options:

Method A: 88% accuracy, 30 ms
Method B: 93% accuracy, 70 ms
Method C: 91% accuracy, 45 ms

Compare the methods and evaluate which should be selected. Justify your choice using constraint-based reasoning.

ANSWER:
Requirements:

- Accuracy > **90%**
- Response time **< 50 ms**

Method	Accuracy	Response Time	Constraint Result
A	88%	30 ms	Accuracy too low
B	93%	70 ms	Response time too high
C	91%	45 ms	Both satisfied

Evaluation

- **Method A:** Fast, but accuracy is only **88%**, so it fails the safety requirement.
- **Method B:** Accuracy is **93%**, but **70 ms** is too slow.
- **Method C:** Accuracy is **91%** and response time is **45 ms**, so it satisfies **both constraints**.

Decision

Method C should be selected. It is the only method that meets both requirements. This makes it the safest and most practical choice for autonomous vehicle detection.

16. Autonomous Navigation Constraint

System must:

- Accuracy ≥ 95%
- Latency < 70 ms

Options:
Model A: 94%, 60 ms
Model B: 96%, 80 ms
Model C: 95%, 65 ms

Compare and evaluate. Justify your decision.

ANSWER:

Requirements:

- Accuracy ≥ **95%**
- Latency **< 70 ms**

Model	Accuracy	Latency	Constraint Result
A	94%	60 ms	Accuracy too low
B	96%	80 ms	Latency too high
C	95%	65 ms	Both satisfied

Evaluation

- **Model A:** Latency is acceptable, but accuracy is only **94%**, below the required 95%.

- **Model B:** Accuracy is **96%**, but latency is **80 ms**, which exceeds the 70 ms limit.
- **Model C:** Accuracy is **95%** and latency is **65 ms**, satisfying both requirements.

Decision

Model C should be selected because it is the **only model that satisfies both constraints**: accuracy ≥ 95% and latency < 70 ms.

17. Industrial Inspection System

Requirements:

- Detect defects with ≥ 95% accuracy
- Real-time operation

Options:
Method A: 92%, fast
Method B: 96%, slow
Method C: 95%, moderate

Compare and evaluate the suitable method. Justify your decision.

ANSWER:

Requirements:

- Defect detection accuracy ≥ **95%**
- **Real-time operation**

Method	Accuracy	Speed	Constraint Result
A	92%	Fast	Accuracy too low
B	96%	Slow	Not suitable for real-time
C	95%	Moderate	Both satisfied

Evaluation

- **Method A:** Fast, but accuracy is only **92%**, so it fails the accuracy requirement.

- **Method B:** Accuracy is **96%**, but it is **slow**, so it may not support real-time inspection.
- **Method C:** Accuracy is exactly **95%** and speed is **moderate**, making it suitable for real-time operation.

Decision

Method C should be selected because it satisfies both requirements: accuracy ≥ 95% and real-time operation. It provides the best practical balance between accuracy and speed.

18. Crowd Monitoring System

Requirements:

- Stable detection in dense crowds
- Moderate accuracy acceptable

Options:
Optical Flow: unstable in dense scenes
Motion Models: stable

Compare and evaluate. Justify your decision.

ANSWER:

Requirements:

- Stable detection in **dense crowds**
- **Moderate accuracy** is acceptable

Method	**Stability in Dense Crowds**	**Evaluation**
Optical Flow	Unstable	Not suitable
Motion Models	Stable	Suitable

Evaluation

- **Optical Flow:** It can detect movement, but becomes **unstable in dense crowds** because many people move close together and their motions overlap.

- **Motion Models:** Provides **stable detection** in crowded environments and is therefore more reliable for this application.

Decision

Motion Models should be selected because stability is the main requirement. Even if its accuracy is only moderate, it is more reliable than Optical Flow in dense crowd conditions.

19. Video Analytics Pipeline

Constraints:

- Accuracy ≥ 90%
- Processing time < 100 ms

Options:
Pipeline A: 88%, 80 ms
Pipeline B: 91%, 120 ms
Pipeline C: 90%, 95 ms

Compare and evaluate the best pipeline. Justify your decision.

ANSWER:

Requirements:

- Accuracy ≥ **90%**
- Processing time **< 100 ms**

Pipeline	Accuracy	Processing Time	Constraint Result
A	88%	80 ms	Accuracy too low
B	91%	120 ms	Processing time too high
C	90%	95 ms	Both satisfied

Evaluation

- **Pipeline A:** Fast at **80 ms**, but accuracy is only **88%**, so it fails the accuracy requirement.

- **Pipeline B:** Has good accuracy (**91%**), but **120 ms** exceeds the 100 ms limit.
- **Pipeline C:** Achieves **90% accuracy** and **95 ms processing time**, satisfying both constraints.

Decision

Pipeline C should be selected because it is the only pipeline that meets both requirements: accuracy ≥ 90% and processing time < 100 ms.

21. Medical Imaging Analysis

Requirements:

- Very high accuracy (>95%)
- Processing time not critical

Options:
Traditional: 85%
Deep Learning: 97%

Compare and evaluate the suitable approach. Justify your decision.

22. Edge Device Vision System

Constraints:

- Low power
- Real-time performance

Options:
Deep Learning: accurate, heavy
Lightweight Model: moderate accuracy, efficient

Compare and evaluate. Justify logically.

ANSWER:

Requirements:

- Accuracy > **95%**
- Processing time is **not critical**

Method	Accuracy	Evaluation
Traditional	85%	Does not meet accuracy requirement
Deep Learning	97%	Meets accuracy requirement

Evaluation

- **Traditional methods:** Achieve only **85% accuracy**, which is below the required **95%**.
- **Deep Learning:** Achieves **97% accuracy**, exceeding the required level. Since processing time is not important, the higher computational requirement is acceptable.

Decision

Deep Learning should be selected because it provides 97% accuracy, which satisfies the very high accuracy requirement. The slower processing is acceptable because processing time is not critical.

29. Autonomous Drone Tracking

Constraints:

- Real-time (<40 ms)
- Accuracy ≥ 85%

Options:
Model A: 83%, 30 ms
Model B: 86%, 50 ms
Model C: 88%, 38 ms

Compare and evaluate. Justify your decision.

ANSWER:

Requirements:

- Real-time response **< 40 ms**
- Accuracy ≥ **85%**

Model Accuracy Response Time Constraint Result

A	83%	30 ms	Accuracy too low
B	86%	50 ms	Response time too high
C	88%	38 ms	Both satisfied

Evaluation

- **Model A:** Fast at **30 ms**, but accuracy is only **83%**, below the required 85%.
- **Model B:** Accuracy is **86%**, but **50 ms** exceeds the 40 ms limit.
- **Model C:** Achieves **88% accuracy** with **38 ms** response time, satisfying both constraints.

Decision

Model C should be selected because it is the **only model that satisfies both requirements**: accuracy ≥ 85% and response time < 40 ms. It provides the best balance of **accuracy and real-time performance**.

Code of Conduct:

I HARITHI S (192421151) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
HARITHI S

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning

Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation